

Cards Under ₹5,000							
Brand Model	XFX HD 6450	Galaxy GT520	Sapphire HD 6450	Forsa GT520	Sparkle GT520	Power-color HD6450	HIS HD 5670
Price (₹)	2,650	2,900	3,000	3,200	3,140	4,180	4,400
Features (Out of 10)	2.6	3.2	3.2	2	2	2	2
Performance (Out of 90)	16.42	11.28	11.03	15.29	10.73	10.28	18.6
Grand Total (Out of 100)	19.02	14.48	14.23	17.29	12.73	12.28	20.6
Specification and Features							
Type of memory / Memory Clock (MHz)	DDR3 / 533	DDR3 / 500	DDR3 / 667	DDR3 / 650	DDR3 / 500	DDR3 / 667	DDR3 / 500
Core / Shader Clock (MHz)	625 / NA	810 / 1620	625 / NA	810 / 1620	810 / 1620	625 / NA	775 / NA
No. of Stream Processors	160	48	160	48	48	160	400
Memory size (MB) / Memory Interface	1024 / 64-bit	1024 / 64-bit	1024 / 64-bit	2048 / 64-bit	1024 / 64-bit	1024 / 64-bit	1024 / 128-bit
Connects on Card	HDMI, DVI, D-Sub	HDMI, DVI, D-Sub	HDMI, DVI, D-Sub	HDMI, DVI, D-Sub	HDMI, DVI, D-Sub	DVI-D, D-Sub, HDMI	DVI-D, D-Sub, Display Port
Accessories Provided	Half Height adapter	2xHalf Height adapter	2xHalf Height adapter	None	None	None	None
Game Provided	N	N	N	N	N	N	N
Synthetic Benchmarks							
3D Mark 11 (Overall)	846	1070	898	1131	1028	911	2009
3D Mark 11 (Graphics)	729	936	774	988	898	788	1814
Unigine Heaven v2.5 (1680x1050, 4xAF, Max Shader, Extreme Tesellation) Score	66	81	75	89	74	74	151
Unigine Heaven v2.5 (1680x1050, 4xAF, Max Shader, Extreme Tesellation) Frame rate	2.6	3.2	3	3.5	3	2.9	6
Temperature (degrees C) (Load)	66	81	73	55	78	74	57
Temperature (degrees C) (Idle)	54	47	45	37	43	41	33
Game Benchmarks							
FarCry 2 (1280x1024, Ultra High, No AA) Avg	12.64	17.01	13.9	18.63	15.7	13.71	26.74
FarCry 2 (1680x1050, Ultra High, No AA) Avg	10.74	14.33	11.8	15.81	13.15	11.62	23.65
FarCry 2 (1920x1080, Ultra High, No AA) Avg	9.65	12.68	10.64	14.11	11.62	10.49	21.49
FarCry 2 (1280x1024, Ultra High, 4xAA) Avg	9.38	12.6	10.57	14.38	11.59	10.44	19.85
FarCry 2 (1680x1050, Ultra High, 4xAA) Avg	7.87	10.44	8.89	11.95	9.61	8.73	16.72
FarCry 2 (1920x1080, Ultra High, 4xAA) Avg	7.08	9.24	7.99	10.89	8.43	7.86	15.14
S.T.A.L.K.E.R.: Call of Pripyat (Day, 1680x1050, No AA, DX11, Ultra Detail) Avg	9.6	14.3	11	18	13.11	10.9	22.5
Resident Evil (1680x1050, 4xAA, Motion Blur on)	12.9	14.7	14.5	16.5	13.6	13.8	27
Resident Evil (1920x1080, 4xAA, Motion Blur on)	11.2	12.9	12.7	14.3	11.9	12.5	23.6
Crysis Warhead (1280x1024, DX10, Med) Avg	8	13	10	14	12	9	17
Crysis Warhead (1680x1050, DX10, Med) Avg	7	10	8	11	10	8	14
Dirt 3 (1280x1024, No AA, Details Medium) Avg	20.59	25.6	22.43	25.04	23.69	22.13	46.37
Dirt 3 (1680x1050, No AA, Details Medium) Avg	16.2	20.89	18.15	22.33	19.81	17.53	36.63
Dirt 3 (1920x1080, No AA, Details Medium) Avg	14.37	18.43	15.92	19.91	17.21	15.35	31.73

marked at full HD resolution to see game performance.

Asus ENGTS450 has an elaborate cooling mechanism. On the top, you have the sturdy black multi-finned passive

heat sink and two heat pipes, forming an inverted U-shaped loop which touches the GPU for faster heat dissipation. On the right-hand side the heat sink extends a good four cm beyond

the printed circuit board of the card. Also, this card is an underclocked version and houses DDR3 memory instead of the GDDR5 that is part of the reference model.

Another interesting GTS 450 card was the Galaxy GTS450 overclocked version whose fan can be detached to minimise the noise. Galaxy had another GTS 450 at stock set-